SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
with specialization in
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

21CSP401L - Major Project / 21CSP402L - Major Project & 21CSP403L - Internship

Batch Number: A8

SECOND REVIEW

Zero-Knowledge Proof Based User Authentication System

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PROBLEM STATEMENT

- Traditional systems rely on password storage
- Password databases vulnerable to breaches
- Hash leaks enable offline brute-force attacks
- Users reuse weak passwords across services
- Need authentication without revealing secrets
- Goal: eliminate password transmission & storage

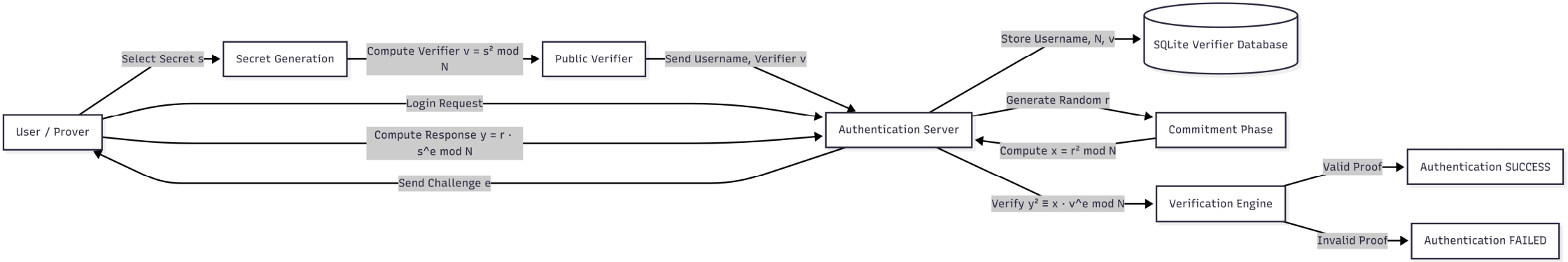
BACKGROUND & MOTIVATION

- Rapid growth of digital platforms increases risk
- Credential theft is a dominant attack vector
- Privacy regulations discourage sensitive storage
- Zero-Knowledge Proofs enable secret-free verification
- Fiat–Shamir protocol widely studied in cryptography
- Motivation: privacy-preserving authentication model

OBJECTIVES

- Design password-less authentication system
- Implement Fiat–Shamir ZKP protocol
- Avoid storing plaintext or hashed passwords
- Demonstrate proof-based identity verification
- Validate success vs failure scenarios
- Analyze security properties of verifier m

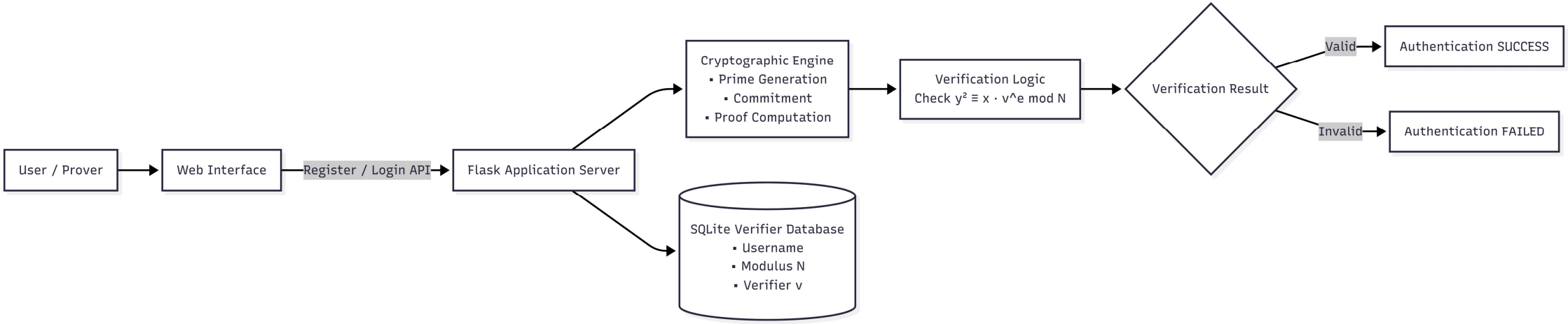
FUNCTIONAL BLOCK DIAGRAM OF THE PROPOSED METHOD



METHODOLOGY

- User selects private secret s
- Public verifier computed: $v = s^2 \bmod N$
- Server stores (username, N, v)
- Login uses challenge–response protocol
- Commitment: random value generation
- Response computed without revealing secret
- Server validates algebraic consistency

ARCHITECTURE DIAGRAM



MATHEMATICAL MODELS & EQUATIONS

Core verifier relation:

- Verifier generation: $v = s^2 \bmod N$
- Commitment: $x = r^2 \bmod N$
- Response: $y = r \cdot s^e \bmod N$
- Verification Equation: $y2 \equiv x*ve(modN)$

Interpretation:

- Valid only if prover knows secret
- No secret disclosure required

Implementation

EXPLORER

- OPEN EDITORS
- ZKP_AUTH_PROJECT
  - templates
    - dashboard.html
    - login.html
    - register.html
  - uploads
    - ~$First Review PPT.pptx
    - First Review PPT.pptx
    - recent_photograph.jpeg
  - app.py
- OUTLINE
- TIMELINE

app.py | dashboard.html

app.py > ...

```python
from flask import Flask, render_template, request, redirect, session, flash
import hashlib
import random
import os

app = Flask(__name__)
app.secret_key = "zkp_secret_key"

users_db = {}

UPLOAD_FOLDER = "uploads"
os.makedirs(UPLOAD_FOLDER, exist_ok=True)

p = 1019
g = 2


def password_to_secret(password):
    hashed = hashlib.sha256(password.encode()).hexdigest()
    return int(hashed, 16) % (p - 1)


def get_user_ip():
    return request.remote_addr


@app.route("/")
def home():
    return redirect("/login")


# --------------- REGISTER ---------------

@app.route("/register", methods=["GET", "POST"])
def register():

    if request.method == "POST":

        username = request.form["username"]
```

Implementation

- Developed a Zero-Knowledge Proof (ZKP) based authentication system that verifies users without revealing their passwords.
- Implemented a secure login mechanism where the server validates cryptographic proofs instead of storing sensitive credentials.
- Designed a Security Dashboard to display user profile, login IP, connected device, and threat level status.
- Integrated secure file storage allowing authenticated users to upload and manage files safely.
- Added IP monitoring and device tracking to detect suspicious login activities.
- Provided a ZKP authentication visualization module to demonstrate the challenge–response verification process.

RESULTS AND DISCUSSION

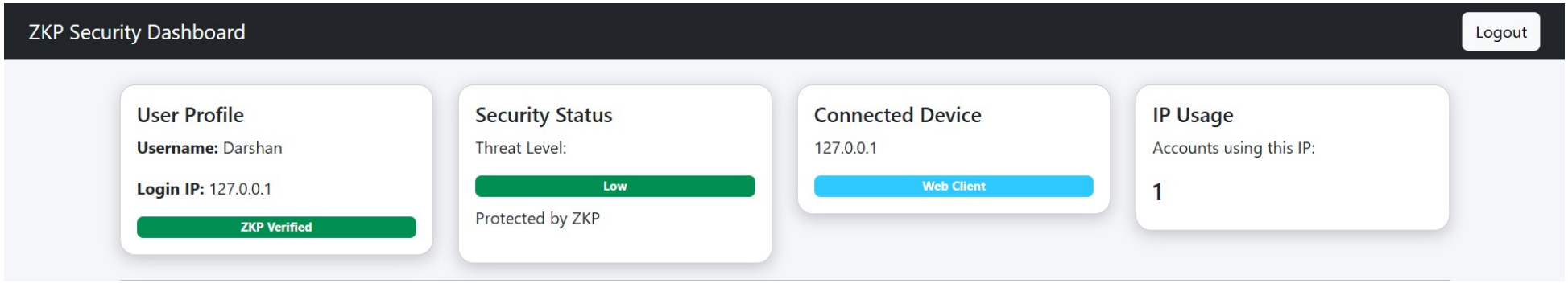
Example Observations:

- Correct secret → Verification SUCCESS
- Incorrect secret → Verification FAILED
- Unknown user → Authentication rejected

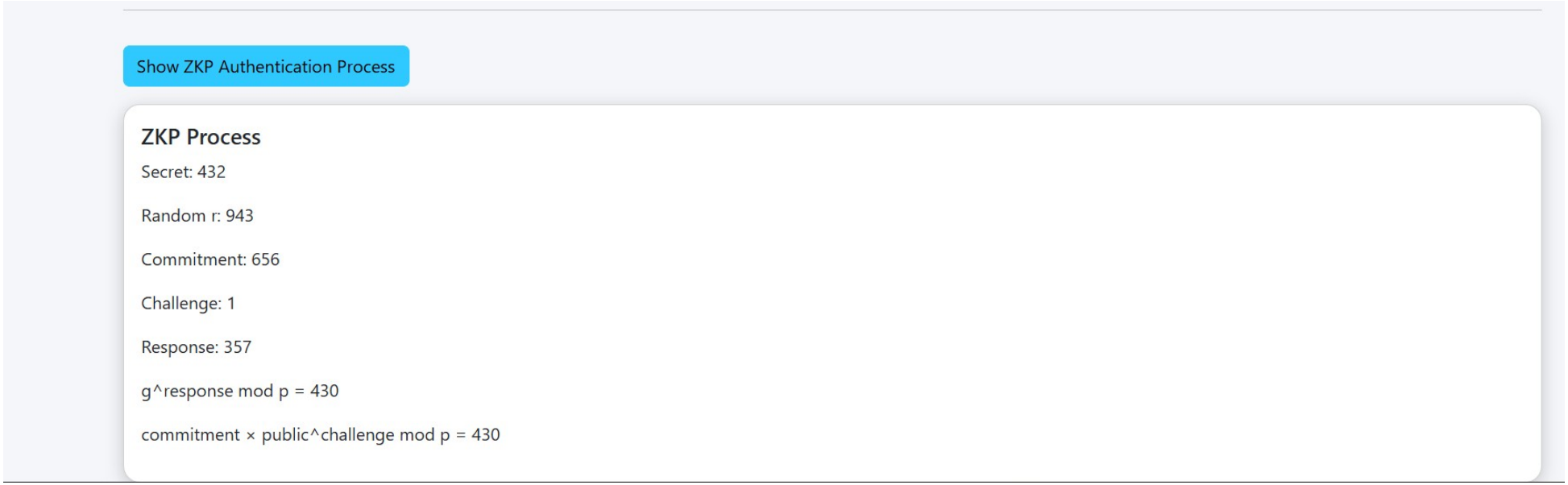
Security implications:

- No password comparison performed
- Proof validated via modular arithmetic

RESULTS AND DISCUSSION



RESULTS AND DISCUSSION



TOOLS TO BE USED

- Programming Language: Python
- Backend Framework: Flask
- Database: SQLite
- Cryptographic Library: PyCryptodome
- Tunneling / Testing: ngrok
- Interface: Browser-based UI

SCOPE OF THE PROJECT

- Privacy-preserving authentication systems
- Secure login for sensitive applications
- Banking / financial identity verification
- High-security distributed systems
- Academic foundation for advanced ZKP systems

Limitations:

- Demonstration-scale parameters
- Single-bit challenge (educational model) Maintain Font size between 20 to 24

REFERENCES

1. • Fiat, A., Shamir, A. (1986) – Identification Scheme
2. • Goldreich, Micali, Wigderson – Zero-Knowledge Proofs
3. • Boneh & Shoup – Applied Cryptography
4. • IEEE Security & Privacy – ZKP Authentication